Band-1 Computational Experiment: Dedekind numbers M(n) — live compute frontier

Human + ZoraASI full-dash (executor)

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Honesty stamp. This note does **not** invent a new exact Dedekind number for large n. Exact values are known through $M(9)$ (independent 2023 computations). The exact compute frontier is **OPEN at** $n = 10$. Mid-2026 Monte Carlo papers give estimates $\widehat{M}(10) \approx 8.93 \times 10^{78}$ only — **not** exact. Tonight's box work certifies classical $M(0) \dots M(6)$ by exhaustive enumeration and corrects a prior finite-ledger mislabel of $M(8)/M(9)$/frontier.

Abstract

The Dedekind number $M(n)$ (OEIS A000372) counts monotone Boolean functions $\{0,1\}^n \to \{0,1\}$, equivalently downsets (or antichains) of the Boolean lattice B_n. This Band-1 probe:

1. Audits 2023–2026 announcements before declaring any n open.
2. Box-certifies $M(0), \dots, M(6)$ by popcount-ordered DFS over monotone assignments.
3. Ledgers published exact $M(7), M(8), M(9)$ with primary citations.
4. Marks $n = 10$ as the next open exact value; records 2026 estimate papers separately.

Bound / value move this run: methods validation + corrected literature frontier (open exact $n = 10$, not $n = 9$). No new astronomical exact $M(n)$.

Problem statement

n	Standing exact $M(n)$	Source class
0	2	Dedekind 1897
1	3	Dedekind 1897
2	6	Dedekind 1897
3	20	Dedekind 1897
4	168	Dedekind 1897
5	7581	Church 1940
6	7828354	Ward 1946
7	2414682040998	Church 1965
8	56130437228687557907788	Wiedemann 1991

n	Standing exact $M(n)$	Source class
9	286386577668298411128469151667598498812366	Jäkel 2023; Van Hirtum et al. 2023/24
10	unknown (exact)	**OPEN** (estimates only, 2026)

Literature ledger (2023–2026 honesty check)

Exact breakthroughs

- $M(8)$ — Wiedemann, *Order* **8** (1991): 56130437228687557907788 (23 digits). Cray-2 era computer-assisted result; still the standing exact value.
- $M(9)$ — independently computed in **April 2023** by two groups unaware of each other (Quanta Magazine, 2023-08-01):
  - Christian Jäkel — GPU / matrix methods; *J. Comput. Algebra* (2023); arXiv:2304.00895.
  - Lennart Van Hirtum, Patrick De Causmaecker, et al. — FPGA supercomputing on Noctua 2; arXiv:2304.03039; ACM *TRETS* **17**(3) (2024).
  - Agreed value: 286386577668298411128469151667598498812366 (42 digits).

Not exact (do not promote to ledger)

- Chen–Feng–Wang–Zhang, arXiv:2606.09795 (2026-06): layer-ratio Monte Carlo $\widehat{M}(10) = (8.9360 \pm 0.0010) \times 10^{78}$.
- arXiv:2607.08446 (2026): statistical estimates $M(10) \ldots M(15)$, e.g. $M(10) \approx 8.93345 \times 10^{78}$ (S.E. 2.44×10^{74}).

Frontier conclusion: exact known for $0 \le n \le 9$; next open exact is $n = 10$.

Errata vs prior `dedekind_finite` stub

A same-day finite stub incorrectly listed a wrong 23-digit string for $M(8)$, attributed $M(8)$ to 2023 authors, and marked $M(9)$ OPEN. **Corrected here:** $M(8)$ = Wiedemann 1991 string above; $M(9)$ = 2023 dual-confirmed value; open exact = 10.

Method (box)

Enumerate monotone Boolean functions by DFS over masks $0 \ldots 2^n - 1$ ordered by popcount. When visiting mask m, if any covered lower neighbor (clear one set bit) is assigned 0, then $f(m)$ is forced to 0; otherwise both 0 and 1 are tried. This enforces monotonicity and counts each function once.

- Code: `/workspace/experiments/band1/dedekind/dedekind_frontier.py` (stdlib only).
- Also retained: earlier `dedekind_finite_ledger.py` (superseded ledger; do not trust its $M(8)$/frontier fields).

Method limits

- Exhaustive certification tonight: $n \le 6$ ($M(6) = 7828354$; wall ≈ 24.3 s).
- $M(7)$ ($\approx 2.4 \times 10^{12}$) and beyond are outside a single-unit naive DFS budget.
- No claim of a new algorithm competitive with Wiedemann / Jäkel / Van Hirtum pipelines.

Results (numbers!)

Box-certified (exact match to literature)

n	Box $M(n)$	Literature	Match	Wall (s)
0	2	2	yes	$< 10^{-4}$
1	3	3	yes	$< 10^{-4}$
2	6	6	yes	$< 10^{-4}$
3	20	20	yes	$< 10^{-4}$
4	168	168	yes	≈ 0.0002
5	7581	7581	yes	≈ 0.020
6	7828354	7828354	yes	≈ 24.30

All seven values match. Total enumeration wall ≈ 24.32 s on Python 3 / Linux box.

Literature exact (not recomputed on box)

n	$M(n)$	Citation
7	2414682040998	Church 1965
8	56130437228687557907788	Wiedemann 1991
9	286386577668298411128469151667598498812366	Van Hirtum et al. 2023/24

Frontier

- **Last exact:** $n = 9$.
- **Next open exact:** $n = 10$.
- **Estimates for $M(10)$:** $\approx 8.93 \times 10^{78}$ (2026 Monte Carlo) — **not** adopted as exact.

Why Band-1

- **Band-1:** classical small-n recomputation, citation hygiene, frontier labeling after checking recent announcements.
- **Not Band-4:** no claim that Dedekind's problem is solved; no invented $M(10)$.

Reproducibility

- Code / data: `/workspace/experiments/band1/dedekind/`
- Run: `python3 dedekind_frontier.py`
- Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`
- Log: `/workspace/experiments/logs/2026-09-11_dedekind.log`
- Merges corrected content from prior `dedekind_finite` stub into this frontier paper.

Conclusion

Box: $M(0) \ldots M(6) = (2, 3, 6, 20, 168, 7581, 7828354)$, all matching literature. **Literature:** exact through $M(9)$ (2023 dual confirmation). **Frontier:** exact $M(10)$ **OPEN**; 2026 papers estimate only. Prior finite-ledger mislabel of open $n = 9$ is withdrawn.